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Parabola

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 69 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let O be the vertex of the parabola $y^2 = 4x$ and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point of the line segment PQ is a conic C , then the length of its latus rectum is :

Open solution

(A) 1 (B) 2 (C) 4 (D) 8

Q2. JEE Main 2026 (Online) 6th April Morning Shift

Let one root of the quadratic equation in x :

$(k^2 - 15k + 27)x^2 + 9(k-1)x + 18 = 0$

be twice the other. Then the length of the latus rectum of the parabola $y^2 = 6kx$ is equal to:

Open solution

(A) 4 (B) 6 (C) 8 (D) 12

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Let chord PQ of length $3\sqrt{13}$ of the parabola $y^2 = 12x$ be such that the ordinates of points P and Q are in the ratio 1:2. If the chord PQ subtends an angle α at the focus of the parabola, then $\sin\alpha$ is equal to :

Open solution

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{5}{13}$ (D) $\frac{12}{13}$

Q4. JEE Main 2026 (Online) 5th April Evening Shift

Let the directrix of the parabola $\mathrm{P} : y^2 = 8x$, cut x-axis at the point A . Let $\mathrm{B}(\alpha, \beta), \alpha > 1$, be a point on P such that the slope of AB is $3/5$. If BC is a focal chord of P, then six times the area of $\triangle ABC$ is :

Open solution

(A) 80 (B) 160 (C) 174 (D) 192

Q5. JEE Main 2026 (Online) 2nd April Evening Shift

Let the parabola $y = x^2 + px + q$ passing through the point $(1, -1)$ be such that the distance between its vertex and the x-axis is minimum. Then the value of $p^2 + q^2$ is :

Open solution

(A) 2 (B) 4 (C) 5 (D) 8

Q6. JEE Main 2026 (Online) 28th January Evening Shift

Let A be the focus of the parabola $y^2 = 8x$. Let the line $y = mx + c$ intersect the parabola at two distinct points B and C. If the centroid of the triangle ABC is $\left(\frac{7}{3}, \frac{4}{3}\right)$, then $(BC)^2$ is equal to :

Open solution

(A) 89 (B) 80 (C) 32 (D) 41

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Q7. JEE Main 2026 (Online) 24th January Evening Shift

Let the image of parabola $x^2 = 4y$, in the line $x - y = 1$ be $(y + a)^2 = b(x - c)$, $a, b, c \in$ N. Then $a + b + c$ is equal to

(A) 12 (B) 8 (C) 6 (D) 4

Open solution

Q8. JEE Main 2026 (Online) 23rd January Evening Shift

An equilateral triangle OAB is inscribed in the parabola $y^2 = 4x$ with the vertex O at the vertex of the parabola. Then the minimum distance of the circle having AB as a diameter from the origin is

(A) $2(8 - 3\sqrt{3})$ (B) $4(6 + \sqrt{3})$ (C) $4(3 - \sqrt{3})$ (D) $2(3 + \sqrt{3})$

Open solution

Q9. JEE Main 2026 (Online) 22nd January Evening Shift

Let the locus of the mid-point of the chord through the origin O of the parabola $y^2 = 4x$ be the curve S . Let P be any point on S . Then the locus of the point, which internally divides OP in the ratio 3 : 1, is :

(A) $2x^2 = 3y$ (B) $2y^2 = 3x$ (C) $3y^2 = 2x$ (D) $3x^2 = 2y$

Open solution

Q10. JEE Main 2026 (Online) 22nd January Morning Shift

If the chord joining the points $\mathrm{P}_1(x_1, y_1)$ and $\mathrm{P}_2(x_2, y_2)$ on the parabola $y^2 = 12x$ subtends a right angle at the vertex of the parabola, then $x_1x_2 - y_1y_2$ is equal to

(A) 280 (B) 288 (C) 292 (D) 284

Open solution

Q11. JEE Main 2026 (Online) 21st January Evening Shift

Let $y^2 = 12x$ be the parabola with its vertex at O. Let P be a point on the parabola and A be a point on the x-axis such that $\angle OPA = 90^\circ$. Then the locus of the centroid of such triangles OPA is:

(A) $y^2 - 4x + 8 = 0$ (B) $y^2 - 2x + 8 = 0$ (C) $y^2 - 9x + 6 = 0$ (D) $y^2 - 6x + 4 = 0$

Open solution

Q12. JEE Main 2026 (Online) 21st January Evening Shift

Let one end of a focal chord of the parabola $y^2 = 16x$ be $(16, 16)$. If $P(\alpha, \beta)$ divides this focal chord internally in the ratio $5 : 2$, then the minimum value of $\alpha + \beta$ is equal to:

(A) 5 (B) 7 (C) 22 (D) 16

Open solution

Q13. JEE Main 2026 (Online) 21st January Morning Shift

Let O be the vertex of the parabola $x^2 = 4y$ and Q be any point on it. Let the locus of the point P , which divides the line segment OQ internally in the ratio $2 : 3$ be the conic C . Then the equation of the chord of C, which is bisected at the point $(1, 2)$, is :

(A) $5x - 4y + 3 = 0$ (B) $x - 2y + 3 = 0$ (C) $4x - 5y + 6 = 0$ (D) $5x - y - 3 = 0$

Open solution

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Q14. JEE Main 2025 (Online) 7th April Morning Shift

Let P be the parabola, whose focus is $(-2, 1)$ and directrix is $2x + y + 2 = 0$. Then the sum of the ordinates of the points on P, whose abscissa is -2, is

(A) $\frac{5}{2}$ (B) $\frac{3}{2}$ (C) $\frac{3}{4}$ (D) $\frac{1}{4}$

Open solution

Q15. JEE Main 2025 (Online) 4th April Evening Shift

A line passing through the point $\mathrm{A}(-2, 0)$, touches the parabola $\mathrm{P} : y^2 = x - 2$ at the point B in the first quadrant. The area, of the region bounded by the line AB, parabola P and the x-axis, is :

(A) 3 (B) $\frac{7}{3}$ (C) $\frac{8}{3}$ (D) 2

Open solution

Q16. JEE Main 2025 (Online) 4th April Evening Shift

The axis of a parabola is the line $y = x$ and its vertex and focus are in the first quadrant at distances $\sqrt{2}$ and $2\sqrt{2}$ units from the origin, respectively. If the point $(1, k)$ lies on the parabola, then a possible value of k is :

(A) 8 (B) 3 (C) 9 (D) 4

Open solution

Q17. JEE Main 2025 (Online) 3rd April Morning Shift

The radius of the smallest circle which touches the parabolas $y = x^2 + 2$ and $x = y^2 + 2$ is

(A) $\frac{7\sqrt{2}}{16}$ (B) $\frac{7\sqrt{2}}{8}$ (C) $\frac{7\sqrt{2}}{2}$ (D) $\frac{7\sqrt{2}}{4}$

Open solution

Q18. JEE Main 2025 (Online) 2nd April Evening Shift

Let the point P of the focal chord PQ of the parabola $y^2 = 16x$ be $(1, -4)$. If the focus of the parabola divides the chord PQ in the ratio $m : n, \gcd(m, n) = 1$, then $m^2 + n^2$ is equal to :

(A) 17 (B) 37 (C) 10 (D) 26

Open solution

Q19. JEE Main 2025 (Online) 2nd April Morning Shift

Let the focal chord PQ of the parabola $y^2 = 4x$ make an angle of 60° with the positive x axis, where P lies in the first quadrant. If the circle, whose one diameter is PS, S being the focus of the parabola, touches the y-axis at the point $(0, \alpha)$, then $5\alpha^2$ is equal to:

(A) 15 (B) 25 (C) 20 (D) 30

Open solution

Q20. JEE Main 2025 (Online) 29th January Morning Shift

Two parabolas have the same focus (4, 3) and their directrices are the x-axis and the y-axis, respectively. If these parabolas intersect at the points A and B, then $(AB)^2$ is equal to :

(A) 384 (B) 392 (C) 96 (D) 192

Open solution

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Q21. JEE Main 2025 (Online) 28th January Morning Shift

Let ABCD be a trapezium whose vertices lie on the parabola $y^2 = 4x$. Let the sides AD and BC of the trapezium be parallel to y-axis. If the diagonal AC is of length $\frac{25}{4}$ and it passes through the point $(1, 0)$, then the area of $ABCD$ is

Open solution

(A) $\frac{75}{8}$ (B) $\frac{125}{8}$ (C) $\frac{25}{2}$ (D) $\frac{75}{4}$

Q22. JEE Main 2025 (Online) 24th January Evening Shift

If the equation of the parabola with vertex $V\left(\frac{3}{2}, 3\right)$ and the directrix $x + 2y = 0$ is $\alpha x^2 + \beta y^2 - \gamma xy - 30x - 60y + 225 = 0$, then $\alpha + \beta + \gamma$ is equal to :

Open solution

(A) 6 (B) 8 (C) 7 (D) 9

Q23. JEE Main 2025 (Online) 23rd January Evening Shift

Let the shortest distance from $(a, 0), a > 0$, to the parabola $y^2 = 4x$ be 4 . Then the equation of the circle passing through the point $(a, 0)$ and the focus of the parabola, and having its centre on the axis of the parabola is :

Open solution

(A) $x^2 + y^2 - 8x + 7 = 0$ (B) $x^2 + y^2 - 6x + 5 = 0$

(C) $x^2 + y^2 - 4x + 3 = 0$ (D) $x^2 + y^2 - 10x + 9 = 0$

Q24. JEE Main 2025 (Online) 23rd January Morning Shift

If the line $3x - 2y + 12 = 0$ intersects the parabola $4y = 3x^2$ at the points A and B, then at the vertex of the parabola, the line segment AB subtends an angle equal to

Open solution

(A) $\frac{\pi}{2} - \tan^{-1}\left(\frac{3}{2}\right)$ (B) $\tan^{-1}\left(\frac{9}{7}\right)$

(C) $\tan^{-1}\left(\frac{11}{9}\right)$ (D) $\tan^{-1}\left(\frac{4}{5}\right)$

Q25. JEE Main 2025 (Online) 22nd January Evening Shift

Let $P(4, 4\sqrt{3})$ be a point on the parabola $y^2 = 4ax$ and PQ be a focal chord of the parabola. If M and N are the foot of perpendiculars drawn from P and Q respectively on the directrix of the parabola, then the area of the quadrilateral PQMN is equal to :

Open solution

(A) $\frac{34\sqrt{3}}{3}$ (B) $\frac{343\sqrt{3}}{8}$ (C) $17\sqrt{3}$ (D) $\frac{263\sqrt{3}}{8}$

Q26. JEE Main 2025 (Online) 22nd January Morning Shift

Let the parabola $y = x^2 + px - 3$, meet the coordinate axes at the points P, Q and R . If the circle C with centre at $(-1, -1)$ passes through the points P, Q and R, then the area of $\triangle PQR$ is :

Open solution

(A) 4 (B) 6 (C) 5 (D) 7

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Q27. JEE Main 2024 (Online) 6th April Morning Shift

Let C be the circle of minimum area touching the parabola $y = 6 - x^2$ and the lines $y = \sqrt{3}|x|$. Then, which one of the following points lies on the circle C ?

(A) $(1, 2)$ (B) $(2, 2)$ (C) $(1, 1)$ (D) $(2, 4)$

Open solution

Q28. JEE Main 2024 (Online) 4th April Evening Shift

Let PQ be a chord of the parabola $y^2 = 12x$ and the midpoint of PQ be at $(4, 1)$. Then, which of the following point lies on the line passing through the points P and Q ?

(A) $(3, -3)$ (B) $\left(\frac{1}{2}, -20\right)$ (C) $(2, -9)$ (D) $\left(\frac{3}{2}, -16\right)$

Open solution

Q29. JEE Main 2024 (Online) 27th January Morning Shift

If the shortest distance of the parabola $y^2 = 4x$ from the centre of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$ is d, then d^2 is equal to :

(A) 16 (B) 24 (C) 20 (D) 36

Open solution

Q30. JEE Main 2023 (Online) 13th April Morning Shift

Let PQ be a focal chord of the parabola $y^2 = 36x$ of length 100 , making an acute angle with the positive x-axis. Let the ordinate of P be positive and M be the point on the line segment PQ such that PM : MQ = 3 : 1. Then which of the following points does NOT lie on the line passing through M and perpendicular to the line PQ?

(A) $(6, 29)$ (B) $(-3, 43)$ (C) $(3, 33)$ (D) $(-6, 45)$

Open solution

Q31. JEE Main 2023 (Online) 8th April Morning Shift

Let R be the focus of the parabola $y^2 = 20x$ and the line $y = mx + c$ intersect the parabola at two points P and Q.

Let the point $G(10, 10)$ be the centroid of the triangle PQR. If $c - m = 6$, then $(PQ)^2$ is :

(A) 317 (B) 325 (C) 346 (D) 296

Open solution

Q32. JEE Main 2023 (Online) 30th January Evening Shift

The parabolas : $ax^2 + 2bx + cy = 0$ and $dx^2 + 2ex + fy = 0$ intersect on the line $y = 1$. If a, b, c, d, e, f are positive real numbers and a, b, c are in G.P., then :

(A) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P. (B) $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in G.P.

(C) d, e, f are in A.P. (D) d, e, f are in G.P.

Open solution

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Q33. JEE Main 2023 (Online) 25th January Evening Shift

The equations of two sides of a variable triangle are $x = 0$ and $y = 3$, and its third side is a tangent to the parabola $y^2 = 6x$. The locus of its circumcentre is :

(A) $4y^2 - 18y - 3x - 18 = 0$
(B) $4y^2 + 18y + 3x + 18 = 0$
(C) $4y^2 - 18y + 3x + 18 = 0$
(D) $4y^2 - 18y - 3x + 18 = 0$

Open solution

Q34. JEE Main 2022 (Online) 29th July Morning Shift

Let the focal chord of the parabola $\mathrm{P} : y^2 = 4x$ along the line $\mathrm{L} : y = \mathrm{m}x + \mathrm{c}, \mathrm{m} > 0$ meet the parabola at the points M and N. Let the line L be a tangent to the hyperbola $\mathrm{H} : x^2 - y^2 = 4$. If O is the vertex of P and F is the focus of H on the positive x-axis, then the area of the quadrilateral OMFN is :

(A) $2\sqrt{6}$
(B) $2\sqrt{14}$
(C) $4\sqrt{6}$
(D) $4\sqrt{14}$

Open solution

Q35. JEE Main 2022 (Online) 26th July Evening Shift

Let P and Q be any points on the curves $(x-1)^2 + (y+1)^2 = 1$ and $y = x^2$, respectively. The distance between P and Q is minimum for some value of the abscissa of P in the interval :

(A) $\left(0, \frac{1}{4}\right)$
(B) $\left(\frac{1}{2}, \frac{3}{4}\right)$
(C) $\left(\frac{1}{4}, \frac{1}{2}\right)$
(D) $\left(\frac{3}{4}, 1\right)$

Open solution

Q36. JEE Main 2022 (Online) 29th June Morning Shift

Let PQ be a focal chord of the parabola $y^2 = 4x$ such that it subtends an angle of $\frac{\pi}{2}$ at the point (3, 0). Let the line segment PQ be also a focal chord of the ellipse $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a^2 > b^2$. If e is the eccentricity of the ellipse E, then the value of $\frac{1}{e^2}$ is equal to :

(A) $1 + \sqrt{2}$
(B) $3 + 2\sqrt{2}$
(C) $1 + 2\sqrt{3}$
(D) $4 + 5\sqrt{3}$

Open solution

Q37. JEE Main 2022 (Online) 28th June Evening Shift

If vertex of a parabola is $(2, -1)$ and the equation of its directrix is $4x - 3y = 21$, then the length of its latus rectum is :

(A) 2
(B) 8
(C) 12
(D) 16

Open solution

Q38. JEE Main 2022 (Online) 27th June Evening Shift

If the equation of the parabola, whose vertex is at (5, 4) and the directrix is $3x + y - 29 = 0$, is $x^2 + ay^2 + bxy + cx + dy + k = 0$, then $a + b + c + d + k$ is equal to :

(A) 575
(B) -575
(C) 576
(D) -576

Open solution

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Q39. JEE Main 2022 (Online) 25th June Morning Shift

Let $x = 2t, y = \frac{t^2}{3}$ be a conic. Let S be the focus and B be the point on the axis of the conic such that $SA \perp BA$, where A is any point on the conic. If k is the ordinate of the centroid of the ΔSAB, then $\lim\limits_{t \to 1} k$ is equal to :

(A) $\frac{17}{18}$ (B) $\frac{19}{18}$ (C) $\frac{11}{18}$ (D) $\frac{13}{18}$

Open solution

Q40. JEE Main 2022 (Online) 24th June Evening Shift

A particle is moving in the xy-plane along a curve C passing through the point (3, 3). The tangent to the curve C at the point P meets the x-axis at Q. If the y-axis bisects the segment PQ, then C is a parabola with :

(A) length of latus rectum 3 (B) length of latus rectum 6

(C) focus $\left(\frac{4}{3}, 0\right)$ (D) focus $\left(0, \frac{3}{4}\right)$

Open solution

Q41. JEE Main 2021 (Online) 31st August Morning Shift

The length of the latus rectum of a parabola, whose vertex and focus are on the positive x-axis at a distance R and S (> R) respectively from the origin, is :

(A) 4(S + R) (B) 2(S $-$ R) (C) 4(S $-$ R) (D) 2(S + R)

Open solution

Q42. JEE Main 2021 (Online) 27th August Evening Shift

If two tangents drawn from a point P to the parabola $y^2 = 16(x - 3)$ are at right angles, then the locus of point P is :

(A) x + 3 = 0 (B) x + 1 = 0 (C) x + 2 = 0 (D) x + 4 = 0

Open solution

Q43. JEE Main 2021 (Online) 20th July Evening Shift

Let P be a variable point on the parabola $y = 4x^2 + 1$. Then, the locus of the mid-point of the point P and the foot of the perpendicular drawn from the point P to the line y = x is :

(A) $(3x - y)^2 + (x - 3y) + 2 = 0$ (B) $2(3x - y)^2 + (x - 3y) + 2 = 0$

(C) $(3x - y)^2 + 2(x - 3y) + 2 = 0$ (D) $2(x - 3y)^2 + (3x - y) + 2 = 0$

Open solution

Q44. JEE Main 2021 (Online) 25th February Evening Shift

The shortest distance between the line x $-$ y = 1 and the curve $x^2 = 2y$ is :

(A) 0 (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Open solution

Q45. JEE Main 2021 (Online) 24th February Evening Shift

If P is a point on the parabola $y = x^2 + 4$ which is closest to the straight line y = 4x $-$ 1, then the co-ordinates of P are :

(A) $(-2, 8)$ (B) (2, 8) (C) (1, 5) (D) (3, 13)

Open solution

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Q46. JEE Main 2021 (Online) 24th February Morning Shift

The locus of the mid-point of the line segment joining the focus of the parabola $y^2 = 4ax$ to a moving point of the parabola, is another parabola whose directrix is :

(A) x = 0 (B) $x = -\frac{a}{2}$ (C) x = a (D) $x = \frac{a}{2}$

Open solution

Q47. JEE Main 2020 (Online) 3rd September Evening Slot

Let the latus rectum of the parabola $y^2 = 4x$ be the common chord to the circles C_1 and C_2 each of them having radius $2\sqrt{5}$. Then, the distance between the centres of the circles C_1 and C_2 is :

(A) 8 (B) 12 (C) $8\sqrt{5}$ (D) $4\sqrt{5}$

Open solution

Q48. JEE Main 2020 (Online) 3rd September Morning Slot

Let P be a point on the parabola, $y^2 = 12x$ and N be the foot of the perpendicular drawn from P on the axis of the parabola. A line is now drawn through the mid-point M of PN, parallel to its axis which meets the parabola at Q. If the y-intercept of the line NQ is $\frac{4}{3}$, then :

(A) MQ = $\frac{1}{3}$ (B) PN = 4 (C) PN = 3 (D) MQ = $\frac{1}{4}$

Open solution

Q49. JEE Main 2020 (Online) 2nd September Evening Slot

The area (in sq. units) of an equilateral triangle inscribed in the parabola $y^2 = 8x$, with one of its vertices on the vertex of this parabola, is :

(A) $256\sqrt{3}$ (B) $64\sqrt{3}$ (C) $128\sqrt{3}$ (D) $192\sqrt{3}$

Open solution

Q50. JEE Main 2020 (Online) 8th January Morning Slot

The locus of a point which divides the line segment joining the point (0, –1) and a point on the parabola, $x^2 = 4y$, internally in the ratio 1 : 2, is :

(A) $9x^2 - 3y = 2$ (B) $4x^2 - 3y = 2$ (C) $x^2 - 3y = 2$ (D) $9x^2 - 12y = 8$

Open solution

Q51. JEE Main 2019 (Online) 9th April Morning Slot

If one end of a focal chord of the parabola, $y^2 = 16x$ is at (1, 4), then the length of this focal chord is :

(A) 24 (B) 20 (C) 25 (D) 22

Open solution

Q52. JEE Main 2019 (Online) 12th January Morning Slot

The maximum area (in sq. units) of a rectangle having its base on the x-axis and its other two vertices on the parabola, $y = 12 - x^2$ such that the rectangle lies inside the parabola, is :

(A) 36 (B) $20\sqrt{2}$ (C) $18\sqrt{3}$ (D) 32

Open solution

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Q53. JEE Main 2019 (Online) 12th January Morning Slot

Let P(4, −4) and Q(9, 6) be two points on the parabola, $y^2 = 4x$ and let X be any point on the arc POQ of this parabola, where O is the vertex of this parabola, such that the area of ΔPXQ is maximum. Then this maximum area (in sq. units) is :

(A) $\frac{625}{4}$ (B) $\frac{125}{4}$ (C) $\frac{75}{2}$ (D) $\frac{125}{2}$

Open solution

Q54. JEE Main 2019 (Online) 11th January Evening Slot

If the area of the triangle whose one vertex is at the vertex of the parabola, $y^2 + 4(x - a^2) = 0$ and the othertwo vertices are the points of intersection of the parabola and y-axis, is 250 sq. units, then a value of 'a' is :

(A) $5\sqrt{5}$ (B) $(10)^{2/3}$ (C) $5\left(2^{1/3}\right)$ (D) 5

Open solution

Q55. JEE Main 2019 (Online) 10th January Evening Slot

The length of the chord of the parabola $x^2 = 4y$ having equation $x - \sqrt{2}y + 4\sqrt{2} = 0$ is -

(A) $8\sqrt{2}$ (B) $6\sqrt{3}$ (C) $3\sqrt{2}$ (D) $2\sqrt{11}$

Open solution

Q56. JEE Main 2019 (Online) 9th January Evening Slot

Let A(4, − 4) and B(9, 6) be points on the parabola, $y^2 = 4x$. Let C be chosen on the arc AOB of the parabola, where O is the origin, such that the area of ΔACB is maximum. Then, the area (in sq. units) of ΔACB, is :

(A) $31\frac{1}{4}$ (B) $30\frac{1}{2}$ (C) 32 (D) $31\frac{3}{4}$

Open solution

Q57. JEE Main 2019 (Online) 9th January Morning Slot

Axis of a parabola lies along x-axis. If its vertex and focus are at distances 2 and 4 respectively from the origin, on the positive x-axis then which of the following points does not lie on it?

(A) $(5, 2\sqrt{6})$ (B) $(6, 4\sqrt{2})$ (C) (8, 6) (D) (4, -4)

Open solution

Q58. JEE Main 2019 (Online) 9th January Morning Slot

If θ denotes the acute angle between the curves, $y = 10 - x^2$ and $y = 2 + x^2$ at a point of their intersection, the $|\tan\theta|$ is equal to :

(A) $\frac{8}{15}$ (B) $\frac{4}{9}$ (C) $\frac{7}{17}$ (D) $\frac{8}{17}$

Open solution

Q59. JEE Main 2018 (Online) 15th April Evening Slot

Tangents drawn from the point $(-8, 0)$ to the parabola $y^2 = 8x$ touch the parabola at P and Q. If F is the focus of the parabola, then the area of the triangle PFQ (in sq. units) is equal to :

(A) 24 (B) 32 (C) 48 (D) 64

Open solution

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Q60. JEE Main 2015 (Online) 11th April Morning Slot

Let PQ be a double ordinate of the parabola, $y^2 = -4x$,
where P lies in the second quadrant.
If R divides PQ in the ratio 2 : 1,
then the locus of R is :

(A) $9y^2 = 4x$ (B) $9y^2 = -4x$ (C) $3y^2 = 2x$ (D) $3y^2 = -2x$

Open solution

Q61. JEE Main 2015 (Offline)

Let O be the vertex and Q be any point on the parabola, $x^2 = 8y$. If the point P divides the line segment OQ internally in the ratio $1 : 3$, then locus of P is :

(A) $y^2 = 2x$ (B) $x^2 = 2y$ (C) $x^2 = y$ (D) $y^2 = x$

Open solution

Q62. JEE Main 2014 (Online) 19th April Morning Slot

A chord is drawn through the focus of the parabola $y^2 = 6x$ such that its distance from the vertex of this parabola is $\frac{\sqrt{5}}{2}$, then its slope can be :

(A) $\frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2}{\sqrt{5}}$ (D) $\frac{2}{\sqrt{3}}$

Open solution

Q63. JEE Main 2014 (Online) 11th April Morning Slot

Let L_1 be the length of the common chord of the curves $x^2 + y^2 = 9$ and $y^2 = 8x$, and L_2 be the length of the latus rectum of $y^2 = 8x$, then :

(A) $L_1 > L_2$ (B) $L_1 = L_2$
(C) $L_1 < L_2$ (D) $\frac{L_1}{L_2} = \sqrt{2}$

Open solution

Q64. AIEEE 2012 (Online) 12th May

The area of triangle formed by the lines joining the vertex of the parabola, $x^2 = 8y$, to the extremities of its latus rectum is

(A) 2 (B) 8 (C) 1 (D) 4

Open solution

Q65. AIEEE 2010

If two tangents drawn from a point P to the parabola $y^2 = 4x$ are at right angles, then the locus of P is

(A) $2x + 1 = 0$ (B) $x = -1$ (C) $2x - 1 = 0$ (D) $x = 1$

Open solution

Q66. AIEEE 2008

A parabola has the origin as its focus and the line $x = 2$ as the directrix. Then the vertex of the parabola is at :

(A) $(0, 2)$ (B) $(1, 0)$ (C) $(0, 1)$ (D) $(2, 0)$

Open solution

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Q67. AIEEE 2006

The locus of the vertices of the family of parabolas $y = \frac{a^3x^2}{3} + \frac{a^2x}{2} - 2a$ is :

(A) $xy = \frac{105}{64}$ (B) $xy = \frac{3}{4}$ (C) $xy = \frac{35}{16}$ (D) $xy = \frac{64}{105}$

Open solution

Q68. AIEEE 2005

Let P be the point $(1, 0)$ and Q a point on the parabola $y^2 = 8x$. The locus of mid point of PQ is :

(A) $y^2 - 4x + 2 = 0$ (B) $y^2 + 4x + 2 = 0$ (C) $x^2 + 4y + 2 = 0$ (D) $x^2 - 4y + 2 = 0$

Open solution

Q69. AIEEE 2004

If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabolas $y^2 = 4ax$ and $x^2 = 4ay$, then :

(A) $d^2 + (3b - 2c)^2 = 0$ (B) $d^2 + (3b + 2c)^2 = 0$

(C) $d^2 + (2b - 3c)^2 = 0$ (D) $d^2 + (2b + 3c)^2 = 0$

Open solution

Numerical 18 questions

Q70. JEE Main 2026 (Online) 4th April Evening Shift

Let A, B and C be the vertices of a variable right angled triangle inscribed in the parabola $y^2 = 16x$. Let the vertex B containing the right angle be $(4, 8)$ and the locus of the centroid of $\triangle ABC$ be a conic C_0. Then three times the length of latus rectum of C_0 is ____

Open solution

Q71. JEE Main 2025 (Online) 8th April Evening Shift

Let r be the radius of the circle, which touches x - axis at point $(a, 0), a < 0$ and the parabola $y^2 = 9x$ at the point $(4, 6)$. Then r is equal to _____.

Open solution

Q72. JEE Main 2025 (Online) 29th January Evening Shift

Let $y^2 = 12x$ be the parabola and S be its focus. Let PQ be a focal chord of the parabola such that $(SP)(SQ) = \frac{147}{4}$. Let C be the circle described taking PQ as a diameter. If the equation of a circle C is $64x^2 + 64y^2 - \alpha x - 64\sqrt{3}y = \beta$, then $\beta - \alpha$ is equal to .

Open solution

Q73. JEE Main 2025 (Online) 28th January Evening Shift

Let A and B be the two points of intersection of the line $y + 5 = 0$ and the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If d denotes the distance between A and B, and a denotes the area of $\triangle SAB$, where S is the focus of the parabola $y^2 = 4x$, then the value of $(a + d)$ is _____.

Open solution

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Q74. JEE Main 2025 (Online) 23rd January Evening Shift

The focus of the parabola $y^2 = 4x + 16$ is the centre of the circle C of radius 5 . If the values of λ, for which C passes through the point of intersection of the lines $3x - y = 0$ and $x + \lambda y = 4$, are λ_1 and $\lambda_2, \lambda_1 < \lambda_2$, then $12\lambda_1 + 29\lambda_2$ is equal to _____ .

Open solution

Q75. JEE Main 2024 (Online) 9th April Evening Shift

Let A, B and C be three points on the parabola $y^2 = 6x$ and let the line segment AB meet the line L through C parallel to the x-axis at the point D. Let M and N respectively be the feet of the perpendiculars from A and B on L. Then $\left(\frac{AM \cdot BN}{CD}\right)^2$ is equal to _____.

Open solution

Q76. JEE Main 2024 (Online) 9th April Evening Shift

Consider the circle $C : x^2 + y^2 = 4$ and the parabola $P : y^2 = 8x$. If the set of all values of α, for which three chords of the circle C on three distinct lines passing through the point $(\alpha, 0)$ are bisected by the parabola P is the interval (p, q), then $(2q - p)^2$ is equal to _____.

Open solution

Q77. JEE Main 2024 (Online) 6th April Morning Shift

Let a conic C pass through the point $(4, -2)$ and $P(x, y), x \geq 3$, be any point on C. Let the slope of the line touching the conic C only at a single point P be half the slope of the line joining the points P and $(3, -5)$. If the focal distance of the point $(7, 1)$ on C is d, then $12d$ equals _____.

Open solution

Q78. JEE Main 2024 (Online) 6th April Morning Shift

Let L_1, L_2 be the lines passing through the point $P(0, 1)$ and touching the parabola $9x^2 + 12x + 18y - 14 = 0$. Let Q and R be the points on the lines L_1 and L_2 such that the $\triangle PQR$ is an isosceles triangle with base QR. If the slopes of the lines QR are m_1 and m_2, then $16\left(m_1^2 + m_2^2\right)$ is equal to _____.

Open solution

Q79. JEE Main 2024 (Online) 5th April Evening Shift

Let a line perpendicular to the line $2x - y = 10$ touch the parabola $y^2 = 4(x - 9)$ at the point P. The distance of the point P from the centre of the circle $x^2 + y^2 - 14x - 8y + 56 = 0$ is _____.

Open solution

Q80. JEE Main 2024 (Online) 5th April Morning Shift

Suppose AB is a focal chord of the parabola $y^2 = 12x$ of length l and slope $\mathrm{m} < \sqrt{3}$. If the distance of the chord AB from the origin is d, then $l\,\mathrm{d}^2$ is equal to _____.

Open solution

Q81. JEE Main 2024 (Online) 4th April Morning Shift

Let the length of the focal chord PQ of the parabola $y^2 = 12x$ be 15 units. If the distance of PQ from the origin is p, then $10\mathrm{p}^2$ is equal to _____.

Open solution

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Q82. JEE Main 2024 (Online) 1st February Morning Shift

Let the line $\mathrm{L}: \sqrt{2}x + y = \alpha$ pass through the point of the intersection P (in the first quadrant) of the circle $x^2 + y^2 = 3$ and the parabola $x^2 = 2y$. Let the line L touch two circles C_1 and C_2 of equal radius $2\sqrt{3}$. If the centres Q_1 and Q_2 of the circles C_1 and C_2 lie on the y-axis, then the square of the area of the triangle $\mathrm{PQ_1Q_2}$ is equal to _____.

Open solution

Q83. JEE Main 2024 (Online) 29th January Evening Shift

Let $P(\alpha, \beta)$ be a point on the parabola $y^2 = 4x$. If P also lies on the chord of the parabola $x^2 = 8y$ whose mid point is $\left(1, \frac{5}{4}\right)$, then $(\alpha - 28)(\beta - 8)$ is equal to _____.

Open solution

Q84. JEE Main 2023 (Online) 1st February Evening Shift

If the x-intercept of a focal chord of the parabola $y^2 = 8x + 4y + 4$ is 3, then the length of this chord is equal to _____.

Open solution

Q85. JEE Main 2022 (Online) 30th June Morning Shift

Let PQ be a focal chord of length 6.25 units of the parabola $y^2 = 4x$. If O is the vertex of the parabola, then 10 times the area (in sq. units) of ΔPOQ is equal to _____.

Open solution

Q86. JEE Main 2022 (Online) 24th June Evening Shift

Let P_1 be a parabola with vertex (3, 2) and focus (4, 4) and P_2 be its mirror image with respect to the line $x + 2y = 6$. Then the directrix of P_2 is $x + 2y =$ _____.

Open solution

Q87. JEE Main 2021 (Online) 20th July Evening Shift

If the point on the curve $y^2 = 6x$, nearest to the point $\left(3, \frac{3}{2}\right)$ is (α, β), then $2(\alpha + \beta)$ is equal to _____.

Open solution

Out of syllabus 74 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q88. JEE Main 2023 (Online) 31st January Morning Shift · MCQ (Single Correct Answer)

Let $y = f(x)$ represent a parabola with focus $\left(-\frac{1}{2}, 0\right)$ and directrix $y = -\frac{1}{2}$. Then

$S = \left\{x \in \mathbb{R} : \tan^{-1}(\sqrt{f(x)}) + \sin^{-1}(\sqrt{f(x)+1}) = \frac{\pi}{2}\right\}$:

(A) is an empty set

(B) contains exactly one element

(C) contains exactly two elements

(D) is an infinite set

Open solution

Q89. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

Let A be a point on the x-axis. Common tangents are drawn from A to the curves $x^2 + y^2 = 8$ and $y^2 = 16x$. If one of these tangents touches the two curves at Q and R, then $(QR)^2$ is equal to :

(A) 76 (B) 81 (C) 64 (D) 72

Open solution

Q90. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

If $\mathrm{P(h, k)}$ be a point on the parabola $x = 4y^2$, which is nearest to the point $\mathrm{Q}(0, 33)$, then the distance of P from the directrix of the parabola $y^2 = 4(x + y)$ is equal to :

(A) 8 (B) 2 (C) 6 (D) 4

Open solution

Q91. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (Single Correct Answer)

If the tangent at a point P on the parabola $y^2 = 3x$ is parallel to the line $x + 2y = 1$ and the tangents at the points Q and R on the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ are perpendicular to the line $x - y = 2$, then the area of the triangle PQR is :

(A) $\frac{9}{\sqrt{5}}$ (B) $3\sqrt{5}$ (C) $5\sqrt{3}$ (D) $\frac{3}{2}\sqrt{5}$

Open solution

Q92. JEE Main 2023 (Online) 25th January Morning Shift · MCQ (Single Correct Answer)

The distance of the point $(6, -2\sqrt{2})$ from the common tangent $\mathrm{y = mx + c, m > 0}$, of the curves $x = 2y^2$ and $x = 1 + y^2$ is :

(A) $\frac{1}{3}$ (B) 5 (C) $\frac{14}{3}$ (D) $5\sqrt{3}$

Open solution

Q93. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

The equations of the sides AB and AC of a triangle ABC are $(\lambda + 1)x + \lambda y = 4$ and $\lambda x + (1 - \lambda)y + \lambda = 0$ respectively. Its vertex A is on the y-axis and its orthocentre is (1, 2). The length of the tangent from the point C to the part of the parabola $y^2 = 6x$ in the first quadrant is :

(A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{6}$

Open solution

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Q94. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

Let a tangent to the curve $y^2 = 24x$ meet the curve $xy = 2$ at the points A and B. Then the mid points of such line segments AB lie on a parabola with the :

(A) length of latus rectum 2
(B) directrix 4x = -3
(C) directrix 4x = 3
(D) length of latus rectum $\frac{3}{2}$

Open solution

Q95. JEE Main 2022 (Online) 30th July Evening Shift · MCQ (Single Correct Answer)

Let the curves $y^2 = kx$ and $xy = -1$ have a common tangent, whose slope is $\frac{1}{2}$. Then k can **NOT** lie in the interval

(A) $(2, 4]$
(B) $(-3, -1)$
(C) $[-4, -3)$
(D) $(1, 3)$

Open solution

Q96. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

If the tangents drawn at the points P and Q on the parabola $y^2 = 2x - 3$ intersect at the point $R(0, 1)$, then the orthocentre of the triangle PQR is :

(A) (0, 1)
(B) (2, -1)
(C) (6, 3)
(D) (2, 1)

Open solution

Q97. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

If the length of the latus rectum of a parabola, whose focus is (a, a) and the tangent at its vertex is $x + y = a$, is 16, then $|a|$ is equal to :

(A) $2\sqrt{2}$
(B) $2\sqrt{3}$
(C) $4\sqrt{2}$
(D) 4

Open solution

Q98. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Let $P(a, b)$ be a point on the parabola $y^2 = 8x$ such that the tangent at P passes through the centre of the circle $x^2 + y^2 - 10x - 14y + 65 = 0$. Let A be the product of all possible values of a and B be the product of all possible values of b. Then the value of $A + B$ is equal to :

(A) 0
(B) 25
(C) 40
(D) 65

Open solution

Q99. JEE Main 2022 (Online) 26th July Evening Shift · MCQ (Single Correct Answer)

The equation of a common tangent to the parabolas $y = x^2$ and $y = -(x - 2)^2$ is

(A) $y = 4(x - 2)$
(B) $y = 4(x - 1)$
(C) $y = 4(x + 1)$
(D) $y = 4(x + 2)$

Open solution

Q100. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

The tangents at the points $A(1, 3)$ and $B(1, -1)$ on the parabola $y^2 - 2x - 2y = 1$ meet at the point P. Then the area (in unit ^{2}) of the triangle PAB is :

(A) 4
(B) 6
(C) 7
(D) 8

Open solution

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Q101. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let P : $y^2 = 4ax$, a > 0 be a parabola with focus S. Let the tangents to the parabola P make an angle of $\frac{\pi}{4}$ with the line y = 3x + 5 touch the parabola P at A and B. Then the value of a for which A, B and S are collinear is :

(A) 8 only (B) 2 only (C) $\frac{1}{4}$ only (D) any a > 0

Open solution

Q102. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let the normal at the point P on the parabola $y^2 = 6x$ pass through the point $(5, -8)$. If the tangent at P to the parabola intersects its directrix at the point Q, then the ordinate of the point Q is :

(A) -3 (B) $-\frac{9}{4}$ (C) $-\frac{5}{2}$ (D) -2

Open solution

Q103. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

If the line $y = 4 + kx$, $k > 0$, is the tangent to the parabola $y = x - x^2$ at the point P and V is the vertex of the parabola, then the slope of the line through P and V is :

(A) $\frac{3}{2}$ (B) $\frac{26}{9}$ (C) $\frac{5}{2}$ (D) $\frac{23}{6}$

Open solution

Q104. JEE Main 2022 (Online) 25th June Morning Shift · MCQ (Single Correct Answer)

If $y = m_1x + c_1$ and $y = m_2x + c_2$, $m_1 \neq m_2$ are two common tangents of circle $x^2 + y^2 = 2$ and parabola $y^2 = x$, then the value of $8|m_1m_2|$ is equal to :

(A) $3 + 4\sqrt{2}$ (B) $-5 + 6\sqrt{2}$ (C) $-4 + 3\sqrt{2}$ (D) $7 + 6\sqrt{2}$

Open solution

Q105. JEE Main 2022 (Online) 24th June Morning Shift · MCQ (Single Correct Answer)

Let $x^2 + y^2 + Ax + By + C = 0$ be a circle passing through (0, 6) and touching the parabola $y = x^2$ at (2, 4). Then A + C is equal to _____.

(A) 16 (B) 88/5 (C) 72 (D) -8

Open solution

Q106. JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

Consider the parabola with vertex $\left(\frac{1}{2}, \frac{3}{4}\right)$ and the directrix $y = \frac{1}{2}$. Let P be the point where the parabola meets the line $x = -\frac{1}{2}$. If the normal to the parabola at P intersects the parabola again at the point Q, then $(PQ)^2$ is equal to :

(A) $\frac{75}{8}$ (B) $\frac{125}{16}$ (C) $\frac{25}{2}$ (D) $\frac{15}{2}$

Open solution

Q107. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

A tangent and a normal are drawn at the point P(2, -4) on the parabola $y^2 = 8x$, which meet the directrix of the parabola at the points A and B respectively. If Q(a, b) is a point such that AQBP is a square, then 2a + b is equal to :

(A) -16 (B) -18 (C) -12 (D) -20

Open solution

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Q108. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Let a parabola P be such that its vertex and focus lie on the positive x-axis at a distance 2 and 4 units from the origin, respectively. If tangents are drawn from O(0, 0) to the parabola P which meet P at S and R, then the area (in sq. units) of ΔSOR is equal to :

(A) $16\sqrt{2}$ (B) 16 (C) 32 (D) $8\sqrt{2}$

Open solution

Q109. JEE Main 2021 (Online) 20th July Morning Shift · MCQ (Single Correct Answer)

Let the tangent to the parabola S : $y^2 = 2x$ at the point P(2, 2) meet the x-axis at Q and normal at it meet the parabola S at the point R. Then the area (in sq. units) of the triangle PQR is equal to :

(A) $\frac{25}{2}$ (B) $\frac{35}{2}$ (C) $\frac{15}{2}$ (D) 25

Open solution

Q110. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

Let L be a tangent line to the parabola $y^2 = 4x - 20$ at (6, 2). If L is also a tangent to the ellipse $\frac{x^2}{2} + \frac{y^2}{b} = 1$, then the value of b is equal to :

(A) 20 (B) 14 (C) 16 (D) 11

Open solution

Q111. JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

Let C be the locus of the mirror image of a point on the parabola $y^2 = 4x$ with respect to the line y = x. Then the equation of tangent to C at P(2, 1) is :

(A) x − y = 1 (B) 2x + y = 5 (C) x + 3y = 5 (D) x + 2y = 4

Open solution

Q112. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

If the three normals drawn to the parabola, $y^2 = 2x$ pass through the point (a, 0) $a \neq 0$, then 'a' must be greater than :

(A) $\frac{1}{2}$ (B) 1 (C) -1 (D) $-\frac{1}{2}$

Open solution

Q113. JEE Main 2021 (Online) 25th February Morning Shift · MCQ (Single Correct Answer)

A tangent is drawn to the parabola $y^2 = 6x$ which is perpendicular to the line 2x + y = 1. Which of the following points does NOT lie on it?

(A) (0, 3) (B) $(-6, 0)$ (C) (4, 5) (D) (5, 4)

Open solution

Q114. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

The centre of the circle passing through the point (0, 1) and touching the parabola $y = x^2$ at the point (2, 4) is :

(A) $\left(\frac{6}{5}, \frac{53}{10}\right)$ (B) $\left(\frac{3}{10}, \frac{16}{5}\right)$

(C) $\left(\frac{-53}{10}, \frac{16}{5}\right)$ (D) $\left(\frac{-16}{5}, \frac{53}{10}\right)$

Open solution

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Q115. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

Let L_1 be a tangent to the parabola $y^2 = 4(x + 1)$
and L_2 be a tangent to the parabola $y^2 = 8(x + 2)$
such that L_1 and L_2 intersect at right angles. Then L_1 and L_2 meet on the straight line :

(A) $x + 3 = 0$ (B) $x + 2y = 0$ (C) $x + 2 = 0$ (D) $2x + 1 = 0$

Open solution

Q116. JEE Main 2020 (Online) 5th September Morning Slot · MCQ (Single Correct Answer)

If the common tangent to the parabolas,
$y^2 = 4x$ and $x^2 = 4y$ also touches the circle, $x^2 + y^2 = c^2$,
then c is equal to :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

Open solution

Q117. JEE Main 2020 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

If one end of a focal chord AB of the parabola $y^2 = 8x$ is at $A\left(\frac{1}{2}, -2\right)$, then the equation of the tangent to it at B is :

(A) $2x - y - 24 = 0$ (B) $x - 2y + 8 = 0$ (C) $x + 2y + 8 = 0$ (D) $2x + y - 24 = 0$

Open solution

Q118. JEE Main 2020 (Online) 7th January Morning Slot · MCQ (Single Correct Answer)

If $y = mx + 4$ is a tangent to both the parabolas, $y^2 = 4x$ and $x^2 = 2by$, then b is equal to :

(A) -128 (B) 128 (C) -64 (D) -32

Open solution

Q119. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

The equation of common tangent to the curves $y^2 = 16x$ and $xy = -4$, is :

(A) $x - y + 4 = 0$ (B) $x + y + 4 = 0$ (C) $x - 2y + 16 = 0$ (D) $2x - y + 2 = 0$

Open solution

Q120. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

The tangents to the curve $y = (x - 2)^2 - 1$ at its points of intersection with the line $x - y = 3$, intersect at the point :

(A) $\left(\frac{5}{2}, -1\right)$ (B) $\left(-\frac{5}{2}, -1\right)$
(C) $\left(\frac{5}{2}, 1\right)$ (D) $\left(-\frac{5}{2}, 1\right)$

Open solution

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Q121. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

Let P be the point of intersection of the common tangents to the parabola $y^2 = 12x$ and the hyperbola $8x^2 - y^2 = 8$. If S and S' denote the foci of the hyperbola where S lies on the positive x-axis then P divides SS' in a ratio :

(A) 14 : 13 (B) 13 : 11 (C) 5 : 4 (D) 2 : 1

Open solution

Q122. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

If the line ax + y = c, touches both the curves $x^2 + y^2 = 1$ and $y^2 = 4\sqrt{2}x$, then |c| is equal to :

(A) 2 (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Open solution

Q123. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

The area (in sq. units) of the smaller of the two circles that touch the parabola, $y^2 = 4x$ at the point (1, 2) and the x-axis is :-

(A) $4\pi\left(3+\sqrt{2}\right)$ (B) $8\pi\left(2-\sqrt{2}\right)$

(C) $8\pi\left(3-2\sqrt{2}\right)$ (D) $4\pi\left(2-\sqrt{2}\right)$

Open solution

Q124. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

The tangent to the parabola $y^2 = 4x$ at the point where it intersects the circle $x^2 + y^2 = 5$ in the first quadrant, passes through the point :

(A) $\left(-\frac{1}{4}, \frac{1}{2}\right)$ (B) $\left(-\frac{1}{3}, \frac{4}{3}\right)$

(C) $\left(\frac{3}{4}, \frac{7}{4}\right)$ (D) $\left(\frac{1}{4}, \frac{3}{4}\right)$

Open solution

Q125. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The shortest distance between the line y = x and the curve $y^2 = x - 2$ is :

(A) $\frac{7}{4\sqrt{2}}$ (B) $\frac{7}{8}$ (C) $\frac{11}{4\sqrt{2}}$ (D) 2

Open solution

Q126. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

The equation of a tangent to the parabola, $x^2 = 8y$, which makes an angle θ with the positive directions of x-axis, is :

(A) $x = y\cot\theta - 2\tan\theta$ (B) $y = x\tan\theta + 2\cot\theta$

(C) $x = y\cot\theta + 2\tan\theta$ (D) $y = x\tan\theta - 2\cot\theta$

Open solution

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Q127. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

If the parabolas $y^2 = 4b(x - c)$ and $y^2 = 8ax$ have a common normal, then which on of the following is a valid choice for the ordered triad (a, b, c)?

(A) (1, 1, 3) (B) (1, 1, 0) (C) $\left(\frac{1}{2}, 2, 0\right)$ (D) $\left(\frac{1}{2}, 2, 3\right)$

Open solution

Q128. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Equation of a common tangent to the circle, $x^2 + y^2 - 6x = 0$ and the parabola, $y^2 = 4x$ is :

(A) $2\sqrt{3}y = 12x + 1$ (B) $\sqrt{3}y = x + 3$ (C) $2\sqrt{3}y = -x - 12$ (D) $\sqrt{3}y = 3x + 1$

Open solution

Q129. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

Let P be a point on the parabola, $x^2 = 4y$. If the distance of P from the center of the circle, $x^2 + y^2 + 6x + 8 = 0$ is minimum, then the equation of the tangent to the parabola at P, is :

(A) $x + 4y - 2 = 0$ (B) $x - y + 3 = 0$ (C) $x + y + 1 = 0$ (D) $x + 2y = 0$

Open solution

Q130. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

Tangent and normal are drawn at P(16, 16) on the parabola $y^2 = 16x$, which intersect the axis of the parabola at A and B, respectively. If C is the centre of the circle through the points P, A and B and $\angle CPB = \theta$, then a value of $\tan\theta$ is :

(A) $\frac{4}{3}$ (B) $\frac{1}{2}$ (C) 2 (D) 3

Open solution

Q131. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

Two parabolas with a common vertex and with axes along x-axis and y-axis, respectively intersect each other in the first quadrant. If the length of the latus rectum of each parabola is 3, then the equation of the common tangent to the two parabolas is :

(A) $4(x + y) + 3 = 0$ (B) $3(x + y) + 4 = 0$

(C) $8(2x + y) + 3 = 0$ (D) $x + 2y + 3 = 0$

Open solution

Q132. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If $y = mx + c$ is the normal at a point on the parabola $y^2 = 8x$ whose focal distance is 8 units, then $|c|$ is equal to :

(A) $2\sqrt{3}$ (B) $8\sqrt{3}$ (C) $10\sqrt{3}$ (D) $16\sqrt{3}$

Open solution

Q133. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

If the common tangents to the parabola, $x^2 = 4y$ and the circle, $x^2 + y^2 = 4$ intersect at the point P, then the distance of P from the origin, is :

(A) $\sqrt{2} + 1$ (B) $2(3 + 2\sqrt{2})$ (C) $2(\sqrt{2} + 1)$ (D) $3 + 2\sqrt{2}$

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Q134. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

P and Q are two distinct points on the parabola, $y^2 = 4x$, with parameters t and t_1 respectively. If the normal at P passes through Q, then the minimum value of t_1^2 is :

(A) 2 (B) 4 (C) 6 (D) 8

Open solution

Q135. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is:

(A) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$ (B) $x^2 + y^2 - 4x + 9y + 18 = 0$

(C) $x^2 + y^2 - 4x + 8y + 12 = 0$ (D) $x^2 + y^2 - x + 4y - 12 = 0$

Open solution

Q136. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

Two tangents are drawn from a point $(-2, -1)$ to the curve, $y^2 = 4x$. If α is the angle between them, then $|\tan\alpha|$ is equal to :

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\sqrt{3}$ (D) 3

Open solution

Q137. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

The slope of the line touching both the parabolas $y^2 = 4x$ and $x^2 = -32y$ is

(A) $\frac{1}{8}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{3}{2}$

Open solution

Q138. JEE Main 2013 (Online) 23th April Morning Slot · MCQ (Single Correct Answer)

The point of intersection of the normals to the parabola $y^2 = 4x$ at the ends of its latus rectum is:

(A) (0, 2) (B) (3, 0) (C) (0, 3) (D) (2, 0)

Open solution

Q139. JEE Main 2013 (Online) 22th April Morning Slot · MCQ (Single Correct Answer)

Statement 1: The line $x - 2y = 2$ meets the parabola, $y^2 + 2x = 0$ only at the point $(-2, -2)$

Statement 2: The line $y = mx - \frac{1}{2m}(m \neq 0)$ is tangent to the parabola, $y^2 = -2x$ at the point $\left(-\frac{1}{2m^2}, -\frac{1}{m}\right)$.

(A) Statement 1 is true; Statement 2 is false;

(B) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation for Statement 1

(C) Statement 1 is false; Statement 2 is true;

(D) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1

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Q140. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

Given : A circle, $2x^2 + 2y^2 = 5$ and a parabola, $y^2 = 4\sqrt{5}x$.
Statement-1 : An equation of a common tangent to these curves is $y = x + \sqrt{5}$.
Statement-2 : If the line, $y = mx + \frac{\sqrt{5}}{m}$ $(m \neq 0)$ is their common tangent, then m satiesfies $m^4 - 3m^2 + 2 = 0$.

(A) Statement-1 is true; Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is true; Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(C) Statement-1 is true; Statement-2 is false.

(D) Statement-1 is false Statement-2 is true.

Open solution

Q141. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

The chord PQ of the parabola $y^2 = x$, where one end P of the chord is at point $(4, -2)$, is perpendicular to the axis of the parabola. Then the slope of the normal at Q is

(A) -4 (B) $-\frac{1}{4}$ (C) 4 (D) $\frac{1}{4}$

Open solution

Q142. AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

The equation of the normal to the parabola, $x^2 = 8y$ at $x = 4$ is

(A) $x + 2y = 0$ (B) $x + y = 2$ (C) $x - 2y = 0$ (D) $x + y = 6$

Open solution

Q143. AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

Statement 1: $y = mx - \frac{1}{m}$ is always a tangent to the parabola, $y^2 = -4x$ for all non-zero values of m.
Statement 2: Every tangent to the parabola, $y^2 = -4x$ will meet its axis at a point whose abscissa is non-negative.

(A) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

(B) Statement 1 is false, Statement 2 is true.

(C) Statement 1 is true, Statement 2 is false.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

Open solution

Q144. AIEEE 2007 · MCQ (Single Correct Answer)

The equation of a tangent to the parabola $y^2 = 8x$ is $y = x + 2$. The point on this line from which the other tangent to the parabola is perpendicular to the given tangent is :

(A) $(2, 4)$ (B) $(-2, 0)$ (C) $(-1, 1)$ (D) $(0, 2)$

Open solution

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Q145. AIEEE 2003 · MCQ (Single Correct Answer)

The normal at the point $(bt_1^2, 2bt_1)$ on a parabola meets the parabola again in the point $(bt_2^2, 2bt_2)$, then :

(A) $t_2 = t_1 + \frac{2}{t_1}$ (B) $t_2 = -t_1 - \frac{2}{t_1}$

(C) $t_2 = -t_1 + \frac{2}{t_1}$ (D) $t_2 = t_1 - \frac{2}{t_1}$

Open solution

Q146. AIEEE 2002 · MCQ (Single Correct Answer)

Two common tangents to the circle $x^2 + y^2 = 2a^2$ and parabola $y^2 = 8ax$ are :

(A) $x = \pm(y + 2a)$ (B) $y = \pm(x + 2a)$ (C) $x = \pm(y + a)$ (D) $y = \pm(x + a)$

Open solution

Q147. JEE Main 2023 (Online) 11th April Evening Shift · Numerical

Let the tangent to the parabola $y^2 = 12x$ at the point $(3, \alpha)$ be perpendicular to the line $2x + 2y = 3$. Then the square of distance of the point $(6, -4)$ from the normal to the hyperbola $\alpha^2 x^2 - 9y^2 = 9\alpha^2$ at its point $(\alpha - 1, \alpha + 2)$ is equal to _____.

Open solution

Q148. JEE Main 2023 (Online) 10th April Morning Shift · Numerical

Let a common tangent to the curves $y^2 = 4x$ and $(x - 4)^2 + y^2 = 16$ touch the curves at the points P and Q. Then $(PQ)^2$ is equal to _____.

Open solution

Q149. JEE Main 2023 (Online) 8th April Evening Shift · Numerical

The ordinates of the points P and Q on the parabola with focus $(3, 0)$ and directrix $x = -3$ are in the ratio $3 : 1$. If $R(\alpha, \beta)$ is the point of intersection of the tangents to the parabola at P and Q, then $\frac{\beta^2}{\alpha}$ is equal to _____.

Open solution

Q150. JEE Main 2023 (Online) 6th April Morning Shift · Numerical

Let the tangent to the curve $x^2 + 2x - 4y + 9 = 0$ at the point $P(1, 3)$ on it meet the y-axis at A. Let the line passing through P and parallel to the line $x - 3y = 6$ meet the parabola $y^2 = 4x$ at B. If B lies on the line $2x - 3y = 8$, then $(AB)^2$ is equal to _____.

Open solution

Q151. JEE Main 2023 (Online) 31st January Evening Shift · Numerical

Let S be the set of all $a \in N$ such that the area of the triangle formed by the tangent at the point P(b, c), b, $c \in \mathbb{N}$, on the parabola $y^2 = 2ax$ and the lines $x = b, y = 0$ is 16 unit^2, then $\sum_{a \in S} a$ is equal to :

Open solution

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Q152. JEE Main 2023 (Online) 29th January Evening Shift · **Numerical**

A triangle is formed by the tangents at the point (2, 2) on the curves $y^2 = 2x$ and $x^2 + y^2 = 4x$, and the line $x + y + 2 = 0$. If r is the radius of its circumcircle, then r^2 is equal to _____.

Open solution

Q153. JEE Main 2022 (Online) 28th July Evening Shift · **Numerical**

Two tangent lines l_1 and l_2 are drawn from the point $(2, 0)$ to the parabola $2y^2 = -x$. If the lines l_1 and l_2 are also tangent to the circle $(x - 5)^2 + y^2 = r$, then 17r is equal to _____.

Open solution

Q154. JEE Main 2022 (Online) 25th July Morning Shift · **Numerical**

The sum of diameters of the circles that touch (i) the parabola $75x^2 = 64(5y - 3)$ at the point $\left(\frac{8}{5}, \frac{6}{5}\right)$ and (ii) the y-axis, is equal to _____.

Open solution

Q155. JEE Main 2022 (Online) 27th June Morning Shift · **Numerical**

A circle of radius 2 unit passes through the vertex and the focus of the parabola $y^2 = 2x$ and touches the parabola $y = \left(x - \frac{1}{4}\right)^2 + \alpha$, where $\alpha > 0$. Then $(4\alpha - 8)^2$ is equal to _____.

Open solution

Q156. JEE Main 2022 (Online) 26th June Morning Shift · **Numerical**

Let the common tangents to the curves $4(x^2 + y^2) = 9$ and $y^2 = 4x$ intersect at the point Q. Let an ellipse, centered at the origin O, has lengths of semi-minor and semi-major axes equal to OQ and 6, respectively. If e and l respectively denote the eccentricity and the length of the latus rectum of this ellipse, then $\frac{l}{e^2}$ is equal to _____.

Open solution

Q157. JEE Main 2021 (Online) 31st August Evening Shift · **Numerical**

A tangent line L is drawn at the point (2, -4) on the parabola $y^2 = 8x$. If the line L is also tangent to the circle $x^2 + y^2 = a$, then 'a' is equal to _____.

Open solution

Q158. JEE Main 2021 (Online) 20th July Morning Shift · **Numerical**

Let $y = mx + c$, $m > 0$ be the focal chord of $y^2 = -64x$, which is tangent to $(x + 10)^2 + y^2 = 4$. Then, the value of $4\sqrt{2}(m + c)$ is equal to _____.

Open solution

Q159. JEE Main 2021 (Online) 25th February Evening Shift · **Numerical**

A line is a common tangent to the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$. If the two points of contact (a, b) and (c, d) are distinct and lie in the first quadrant, then 2(a + c) is equal to _____.

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Q160. JEE Main 2020 (Online) 3rd September Evening Slot · **Numerical**

If the tangent to the curve, $y = e^x$ at a point (c, e^c) and the normal to the parabola, $y^2 = 4x$ at the point $(1, 2)$ intersect at the same point on the x-axis, then the value of c is _____ .

Open solution

Q161. JEE Main 2020 (Online) 8th January Evening Slot · **Numerical**

Let a line $y = mx$ $(m > 0)$ intersect the parabola, $y^2 = x$ at a point P, other than the origin. Let the tangent to it at P meet the x-axis at the point Q. If area $(\Delta OPQ) = 4$ sq. units, then m is equal to _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **D**	3. **A**	4. **B**	5. **B**
6. **B**	7. **C**	8. **C**	9. **B**	10. **B**
11. **B**	12. **B**	13. **A**	14. **B**	15. **C**
16. **C**	17. **B**	18. **A**	19. **A**	20. **D**
21. **D**	22. **D**	23. **B**	24. **B**	25. **B**
26. **B**	27. **D**	28. **B**	29. **C**	30. **B**
31. **B**	32. **A**	33. **C**	34. **B**	35. **C**
36. **B**	37. **B**	38. **D**	39. **D**	40. **A**
41. **C**	42. **B**	43. **B**	44. **B**	45. **B**
46. **A**	47. **A**	48. **D**	49. **D**	50. **D**
51. **C**	52. **D**	53. **B**	54. **D**	55. **B**
56. **A**	57. **C**	58. **A**	59. **C**	60. **B**
61. **B**	62. **A**	63. **C**	64. **B**	65. **B**
66. **B**	67. **A**	68. **A**	69. **D**	

Numerical

70. **16**	71. **30**	72. **1328**	73. **14**	74. **15**
75. **36**	76. **80**	77. **75**	78. **68**	79. **10**
80. **108**	81. **72**	82. **72**	83. **192**	84. **16**
85. **25**	86. **10**	87. **9**		

Out of syllabus

88. **C**	89. **D**	90. **C**	91. **B**	92. **B**
93. **B**	94. **C**	95. **C**	96. **B**	97. **C**
98. **D**	99. **B**	100. **D**	101. **D**	102. **B**
103. **C**	104. **C**	105. **A**	106. **B**	107. **A**
108. **B**	109. **A**	110. **B**	111. **A**	112. **B**
113. **D**	114. **D**	115. **A**	116. **A**	117. **B**
118. **A**	119. **A**	120. **A**	121. **C**	122. **B**
123. **C**	124. **C**	125. **A**	126. **C**	127. **A**
128. **B**	129. **C**	130. **C**	131. **A**	132. **C**

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133. **B**	134. **D**	135. **C**	136. **D**	137. **C**
138. **B**	139. **B**	140. **B**	141. **A**	142. **D**
143. **D**	144. **B**	145. **B**	146. **B**	147. **116**
148. **32**	149. **16**	150. **292**	151. **146**	152. **10**
153. **9**	154. **10**	155. **63**	156. **4**	157. **2**
158. **34**	159. **9**	160. **4**	161. **0.5**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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